MTH1004

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

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Probability, Statistics and Data

Module Convener: Dr Daniel Williamson

You have 24 hours to complete this paper from the time of its release
Intended duration 2 hours

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Graph Paper and Statistical Tables (Murdoch & Barnes) will be provided. Candidates are permitted to use approved portable electronic calculators in this examination.

This is an **OPEN BOOK** examination

SECTION A

1. (a) Suppose events A and B are such that $P(A), P(B) > 0$.

(i) State the definition of conditional probability.

Naming **every** result that you use:

(ii) Show that

$$P(B^c \mid A) = \frac{P(A) - P(B \cap A)}{P(A)}.$$

(iii) Hence show that

$$P(B^c \mid A) = \frac{P(A \cup B) - P(B)}{P(A)}.$$

(iv) Hence show that

$$P(B^c \mid A) = \frac{1 - P(A^c \cap B^c) - P(B)}{P(A)}.$$

(14)

(b) Suppose that the discrete random variable, X, takes values in the set $S = \{2, 4, 6, 8\}$ and that $P(X = j) = c/j$ for every $j \in S$.

(i) Find c.

(ii) Find $F(3)$ and $F(4)$ where $F(X)$ denotes the cdf of X.

(iii) Find $\mathrm{E}[X]$.

(11)

(c) Assume a baby's usual night-time sleep duration (in hours) is distributed as $N(9, 1.5^2)$.

(i) Use statistical tables to estimate the probability that the baby will sleep for 10 or more hours.

(ii) An infant pain relief medication is thought to have sleep prolonging properties. In order to test this, the medication was administered to 36 babies and their sleep duration measured. On average, their sleep duration was found to be 9.5 hours, with the same standard deviation as before. Conduct a two-sided hypothesis test of size 2.5% comparing the usual sleep duration with the medicated sleep duration. State your null and alternative hypotheses, your test statistic and its null distribution, the critical region, and your conclusion.

(iii) Assume that the average sleep duration and standard deviation remain at 9.5 and 1.5 respectively. What is the minimum sample size that yields a statistically significant result using a two-sided, one sample test and assuming a test of size 2.5%. (16)

(d) The times to failure (in months) of 15 randomly selected mechanical components produced by the same factory had sample mean of 25.00 and sample variance of 345.37. Consider modelling this data set as a realisation of i.i.d. random variables $Y_1, \ldots, Y_{15}$, each following the Exponential distribution:

$$f(y;\theta) = \theta e^{-\theta y} \qquad \text{for } y \geq 0,$$

where $\theta > 0$ is a parameter.

(i) Find $\mathrm{E}[Y_i]$ and $\mathrm{Var}[Y_i]$.

(ii) Find the method of moments estimate for θ. (9)

[**50**]

SECTION B

2. The joint probability density function of two random variables (X and Y) is given by

$$f_{X,Y}(x,y) = \begin{cases} C\sqrt{y}\,(y^{\alpha+1}) \exp\left\{-\frac{y(2\beta+x^2)}{2}\right\}, & x \in (-\infty,\infty), \quad y \in [0,\infty), \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find C. (13)

(b) Find the marginal density of Y. What type of distribution does Y follow? (5)

(c) Find the conditional density of $X \mid Y$. What type of distribution is this? (7)

[25]

3. (a) Let $X \sim \text{Unif}(0,1)$ and $Y = -\log X$. Find the cdf of Y. (5)

(b) It is known that the time people spend reading individual news articles on a website is not normally distributed. A company decided to explore the effect of adding images to news articles on reading time. They create two versions of the same news article, one with images and one without and randomly assigned the next 100 readers to one of those pages. They recorded the reading times (in seconds) spent on those pages. The results are provided below.

Group	N	Mean (seconds)	Standard Deviation
Images	50	25	12
No images	50	20	10

(i) Conduct an appropriate hypothesis test of size 5% of the data to explore the effect of adding images to news articles. State your null and alternative hypotheses, your test statistic and it's null distribution, the critical region, the p-value and your conclusion.

(ii) Re-run the analysis, but this time imagine that we know the reading times are normally distributed with known standard deviation 15. State your test statistic and calculate the p-value. What do you conclude now?

(20)

[25]

4. Andy, Ben and Chris are the only suspects in a crime and police are certain that one and only one of them did it. Police don't think, at this stage, any one of them is more likely than the other to have done it. When a suspect is innocent of the crime, the police know that they have an alibi 90% of the time. Guilty suspects are also often able to arrange alibis, and the police know that this occurs for 40% of guilty suspects. Andy, Ben and Chris don't know each other and *given the true criminal*, the existence of alibis for them is independent.

   (a) Early on in the investigation it emerges that Andy has an alibi. What is the probability that Ben committed the crime? Ensure you write your answer as a proper mathematical argument. (13)

   (b) Later in the investigation, it emerges that Chris also has an alibi. Now what is the probability that Ben committed the crime? *Hint: You may use the idea that if events X and Y are conditionally independent given Z then $P(X \cap Y \mid Z) = P(X \mid Z)P(Y \mid Z)$.* (6)

   (c) It is now discovered that Ben does not have an alibi. Now what is the probability that he committed the crime? (6)

[**25**]